

JAVA-CO1-AT3

1.

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

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CO1-AT3-Coding Challenge

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← Back

QUESTIONS**Q1**
Smart University Payroll Management System
25 marks**Q2**
Smart Vehicle Insurance Claim System
25 marks

2/2 answered

Q1 Smart University Payroll Management System 25 marks

A university employs three categories of staff: Teaching Faculty, Technical Staff, and Visiting Faculty. The payroll system should calculate the monthly salary based on the employee type while maintaining a common employee hierarchy. Requirements Create a superclass Employee with the following private data members: employeeId employeeName department basicSalary Provide: Default constructor Parameterized constructor Getter and Setter methods displayEmployee() calculateSalary() Create the following subclasses: TeachingFaculty teachingHours incentivePerHour Salary: basicSalary + (teachingHours × incentivePerHour) TechnicalStaff overtimeHours overtimeRate Salary: basicSalary + (overtimeHours × overtimeRate) VisitingFaculty lecturesHandled paymentPerLecture Salary: lecturesHandled × paymentPerLecture Override calculateSalary() in all subclasses. Create an array of Employee objects and demonstrate runtime polymorphism. Display: Employee details Monthly salary Highest-paid employee Average salary Search an employee using the employee ID.

Test Case 1 (Normal Case)

Input

3

TeachingFaculty

101

Dr. Arun

OUTPUT:

Your Code

```
1 import java.util.*;
2
3 class Main
4 {
5     public static String readInput(Scanner sc)
6     {
7         String s = "";
8
9         while(sc.hasNextLine())
10        {
11            s = sc.nextLine().trim();
12
13            if(!s.isEmpty())
14            {
15                return s;
16            }
17        }
18
19        return s;
20    }
21
22    public static void main(String args[])
23    {
24        Scanner sc = new Scanner(System.in);
```

Input

```
3
TeachingFaculty
101
Dr. Arun
```

Output

```
Employee Details
Employee ID : 101
Name : Dr. Arun
Department : CSE
Monthly Salary : 60000.0
```

```

22     public static void main(String args[])
23     {
24         Scanner sc = new Scanner(System.in);
25
26         int n = Integer.parseInt(readInput(sc));
27
28         Employee emp[] = new Employee[n];
29
30         for(int i=0;i<n;i++)
31         {
32             String type = readInput(sc);
33
34             int id = Integer.parseInt(readInput(sc));
35             String name = readInput(sc);
36             String department = readInput(sc);
37             double basicSalary = Double.parseDouble(readInput(sc));
38
39             if(type.equalsIgnoreCase("TeachingFaculty"))
40             {
41                 int teachingHours = Integer.parseInt(readInput(sc));
42                 double incentivePerHour = Double.parseDouble(readInput(sc));
43
44                 emp[i] = new TeachingFaculty(id,name,department,
45                     basicSalary,teachingHours,incentivePerHour);
46             }
47             else if(type.equalsIgnoreCase("TechnicalStaff"))

```

```

242     private double paymentPerLecture;
243
244     public VisitingFaculty()
245     {
246     }
247
248
249     public VisitingFaculty(int employeeId,String employeeName,
250         String department,double basicSalary,
251         int lecturesHandled,double paymentPerLecture)
252     {
253         super(employeeId,employeeName,department,basicSalary);
254         this.lecturesHandled = lecturesHandled;
255         this.paymentPerLecture = paymentPerLecture;
256     }
257
258     @Override
259     public double calculateSalary()
260     {
261         return getBasicSalary() + (lecturesHandled * paymentPerLecture);
262     }
263 }

```

✔ Submitted

2.



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QUESTIONS

Q1
Smart University Payroll Management System
25 marks

Q2
Smart Vehicle Insurance Claim System
25 marks

2/2 answered

Q2 Smart Vehicle Insurance Claim System
25 marks

Problem Statement An insurance company processes claims for three categories of vehicles: Car, Bike, and Truck. The claim amount depends on the vehicle type. Requirements Superclass: Vehicle Private members: vehicleNumber ownerName insuredAmount Subclasses: Car Bike Truck Each subclass overrides calculateClaimAmount() using different formulas. The application should: Read details for N vehicles Store them using an array of superclass references Calculate claim amounts using runtime polymorphism Search by vehicle number Display the vehicle with the highest claim amount Question 3 (Very Hard) Online Shopping Order Processing System (50 Marks) Problem Statement An e-commerce company sells products in three categories: Electronics Clothing Grocery The billing process varies for each category. Requirements Superclass: Product Private members: productId productName price quantity Subclasses: Electronics Clothing Grocery Each subclass overrides calculateBill(). The program should: Store products in an array of superclass references Calculate the final bill using runtime polymorphism Search products by ID Display the most expensive order Compute total revenue

Claim Formula (Recommended)

Car

Claim = insuredAmount × 0.80

Bike

OUTPUT:

Claim Amount : 60000.0

Your Code

```
1 import java.util.*;
2
3 class Main
4 {
5     public static String readInput(Scanner sc)
6     {
7         String s="";
8
9         while(sc.hasNextLine())
10        {
11            s=sc.nextLine().trim();
12
13            if(!s.isEmpty())
14                return s;
15        }
16
17        return s;
18    }
19
20    public static void main(String args[])
21    {
22        Scanner sc=new Scanner(System.in);
```

Input
3
Car
TN10AB1234

Output
Vehicle : TN10AB1234
Claim Amount : 640000.0
Vehicle : TN09XY5678
Claim Amount : 60000.0

```

24     int n=Integer.parseInt(readInput(sc));
25
26     Vehicle vehicle[]=new Vehicle[n];
27
28     for(int i=0;i<n;i++)
29     {
30         String type=readInput(sc);
31
32         String vehicleNumber=readInput(sc);
33         String ownerName=readInput(sc);
34         double insuredAmount=Double.parseDouble(readInput(sc));
35
36         if(type.equalsIgnoreCase("Car"))
37         {
38             vehicle[i]=new Car(vehicleNumber,ownerName,insuredA
39         }
40         else if(type.equalsIgnoreCase("Bike"))
41         {
42             vehicle[i]=new Bike(vehicleNumber,ownerName,insured
43         }
44         else if(type.equalsIgnoreCase("Truck"))
45         {
46             vehicle[i]=new Truck(vehicleNumber,ownerName,insure
47         }
48     }
49

```

```

177 class Truck extends Vehicle
178 {
179     public Truck()
180     {
181     }
182 }
183
184 public Truck(String vehicleNumber,String ownerName,double insur
185 {
186     super(vehicleNumber,ownerName,insuredAmount);
187 }
188
189 @Override
190 public double calculateClaimAmount()
191 {
192     return getInsuredAmount() * 0.90;
193 }
194 }

```

 **Submitted**